0.0.1 Interacting Multipoles of Rank 1 and 4

$$+\frac{1}{105}\cdot T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}^{B}\Phi_{\beta\gamma\delta\epsilon}^{A}=R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix}
-945\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\gamma}\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\delta}\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\epsilon}\cdot\delta\left(\gamma,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\gamma}\cdot R_{\delta}\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\gamma}\cdot R_{\epsilon}\cdot\delta\left(\beta,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\alpha}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\beta,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\delta}\cdot\delta\left(\alpha,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\beta}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^{2}\cdot R_{\gamma}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\alpha}\cdot\delta\left(\beta,\gamma\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\alpha}\cdot\delta\left(\beta,\delta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\alpha}\cdot\delta\left(\beta,\epsilon\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\beta}\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\beta}\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\beta}\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\gamma}\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\delta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\gamma}\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\gamma}\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\beta,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\delta}\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\delta}\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\epsilon\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\delta}\cdot\delta\left(\alpha,\epsilon\right)\cdot\delta\left(\beta,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\beta\right)\cdot\delta\left(\gamma,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\gamma\right)\cdot\delta\left(\beta,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^{4}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\delta\right)\cdot\delta\left(\beta,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}
\end{bmatrix}\tag{1}$$

Simplify deltas:

$$
+\frac{1}{105}\cdot T_{\alpha\beta\gamma\delta\epsilon}\mu_{\alpha}^{B}\Phi_{\beta\gamma\delta\epsilon}^{A} = R_z^{-11}\cdot\frac{1}{105}\cdot\begin{bmatrix}
-945\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\gamma}\cdot\delta\left(\delta,\delta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\delta\delta}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\delta}\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\gamma\delta}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\beta}\cdot R_{\epsilon}\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\gamma\gamma\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\gamma}\cdot R_{\delta}\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\gamma\delta}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\gamma}\cdot R_{\epsilon}\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\gamma\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\alpha}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\delta\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\delta}\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\gamma\delta}^{A}\\
+105\cdot R_z^2\cdot R_{\beta}\cdot R_{\gamma}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\gamma\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\beta}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\delta\epsilon}^{A}\\
+105\cdot R_z^2\cdot R_{\gamma}\cdot R_{\delta}\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\gamma\delta\epsilon}^{A}\\
-15\cdot R_z^4\cdot R_{\alpha}\cdot\delta\left(\beta,\beta\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\alpha}\cdot\delta\left(\beta,\beta\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\alpha}\cdot\delta\left(\beta,\beta\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\beta\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\beta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\beta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\beta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\gamma,\gamma\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\gamma\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\gamma}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\gamma}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\gamma}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\gamma}^{A}\\
-15\cdot R_z^4\cdot R_{\delta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\delta}^{A}\\
-15\cdot R_z^4\cdot R_{\delta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\delta}^{A}\\
-15\cdot R_z^4\cdot R_{\delta}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\delta}^{A}\\
-15\cdot R_z^4\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\epsilon}^{A}\\
-15\cdot R_z^4\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\epsilon}^{A}\\
-15\cdot R_z^4\cdot R_{\epsilon}\cdot\delta\left(\alpha,\alpha\right)\cdot\delta\left(\beta,\beta\right)\cdot\mu_{\alpha}^{B}\cdot\Phi_{\alpha\beta\beta\epsilon}^{A}
\end{bmatrix} \tag{2a}
$$

$$
\xrightarrow[]{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\delta\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\delta\epsilon}^A \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \end{bmatrix} \tag{2b}
$$

$$
\xRightarrow{\text{reduced indices}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix}
-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
+105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
-15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
-15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A
\end{bmatrix} \tag{2c}
$$

$$
\xRightarrow{\text{added up}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix}
-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
+210 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
+315 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
+420 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
-45 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
-45 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
-135 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A
\end{bmatrix} \tag{2d}
$$

Simplify R:

$$
+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_\alpha^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix}
-945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zzzz}^A \\
+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\delta\delta}^A \\
+210 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\gamma\gamma}^A \\
+315 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\beta\beta}^A \\
+420 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A \\
-45 \cdot R_z^4 \cdot R_z \cdot \mu_z^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
-45 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
-135 \cdot R_z^4 \cdot R_z \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A
\end{bmatrix} \tag{3a}
$$

$$
\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix}
+420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 45 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
-135 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A
\end{bmatrix} \tag{3b}
$$

$$
\xRightarrow{\text{reduced indices}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix}
+420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 45 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A \\
-135 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A
\end{bmatrix} \tag{3c}
$$

$$\xRightarrow{\text{added up}} R_z^{-11} \cdot \frac{1}{105} \cdot \Big[\ +420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 180 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A\ \Big] \tag{3d}$$

Simplify Dipoles:

$$+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot \Big[\ +420 \cdot R_z^5 \cdot \mu_y^B \cdot \Phi_{yzzz}^A - 180 \cdot R_z^5 \cdot \mu_y^B \cdot \Phi_{yz\beta\beta}^A\ \Big] \tag{4a}$$

$$\xRightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot [+0] \tag{4b}$$

Combining everything:

$$+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon}\mu_\alpha^B \Phi_{\beta\gamma\delta\epsilon}^A = +0 \tag{5}$$